Gene Target Validation Dossier

Target gene: TRPC6 **Disease:** Focal Segmental Glomerulosclerosis **Generated:** 2026-10-05 07:35 UTC
Evidence: 129 kept / 72 dropped

Full evidence report: full_report.md — categorized kept sources with external links

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Executive Summary

✅ **Overall consensus: Support** (confidence 85%) | **Suitability score: 85/100**

Lens Summary

- **Genetics lens** ✅: Support (90%) — Genetic evidence strongly supports TRPC6 as a causal driver of focal segmental glomerulosclerosis (FSGS), with multiple pathogenic and likely pathogenic variants, strong Mendelian validation, and consistent missense-predominant spectrum in a LoF-tolerant gene.
- **Biology lens** ✅: Support (90%) — TRPC6 is strongly supported as a mechanistically understood and druggable target for FSGS, with multiple lines of evidence supporting its role in glomerular function and disease pathogenesis. The protein's subcellular localization to the cell membrane makes it amenable to both small-molecule and antibody-based therapies. CRISPR data indicates dependency across some cancer cell lines, which is not a concern for this non-oncology indication.
- **Clinical lens** ✅: Support (80%) — TRPC6 has strong clinical precedent with multiple Phase 2 and ongoing Phase 3 trials for focal segmental glomerulosclerosis (FSGS). BI 764198, a TRPC6 inhibitor, is being tested in these trials, with one completed Phase 2 trial showing proteinuria responses. The literature supports the role of TRPC6 mutations in causing FSGS and suggests that modulating this target could be therapeutically beneficial.
- **Safety lens** ✅: Support (80%) — The target TRPC6 is relatively LoF-tolerant and has multiple pathogenic variants associated with Focal Segmental Glomerulosclerosis, supporting its relevance to the disease. Mouse KO phenotypes include increased vasoconstriction and systemic arterial blood pressure, which are on-target liabilities in the kidney. Expression levels are high in lung, esophagus muscularis, thyroid, and esophagus gastroesophageal junction, raising concerns for on-target effects in these tissues. However, no serious adverse events or clinical trial data are reported, and the LOEUF indicates tolerance to haploinsufficiency. [⚠ CONSTRAINT GUARD: The text above mentions haploinsufficiency, but LOEUF=0.759 (relatively LoF-tolerant (NOT haploinsufficient)) does not support this — LOEUF ≥ 0.35 rules out haploinsufficiency. This claim should be removed.]
- **Commercial lens** ⚖️: Neutral (70%) — The IP landscape is moderately encumbered by 12 patents naming TRPC6 as a target, though their scope and jurisdiction need further assessment. The competitive field is underserved at the target level but crowded in the broader indication space with one approved drug and multiple active clinical programs. The market size for hereditary steroid-resistant nephrotic syndrome, associated with Focal Segmental Glomerulosclerosis, could not be sized from Orphanet due to lack of validated prevalence data. [⚠ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]

- **Regulatory lens** ❓: Insufficient evidence (0%) — No evidence of this lens's type passed screening; verdict reflects the evidence gap, not a negative finding.

> **Evidence note:** Proceed as the completeness scores for genetics and biology are sufficient, and there are no critical gaps.

Evidence Summary

129 kept / 72 dropped (total: 201)

Kept Evidence

Literature (97)

Source	Detail	Quality	Year	First Author
PMID:42041578	An Inducible hiPSC-Derived Human Podocyte Model for Functional Analysis of TRPC6 Variants Associated with FSGS.	★★★	2026	Batool L
PMID:41481124	Podocyte-specific Translational Profiling In Vivo Uncovers Distinct Patterns in Trpc6-Deficient Podocytes.	★★★	2026	Einloft J
PMID:41276089	TRPC6 inhibition by Z3571: A structure-based strategy to ameliorate glomerular and tubular dysfunction in chronic kidney disease.	★★★	2026	Zhou L
PMID:41100809	Structural and functional insights of the podocyte slit diaphragm complex.	★★★	2026	Qadri AH
PMID:40719843	Targeting TRPC6 in podocytopathies: Why clinical translation remains a challenge?	★★★	2026	Patel CA
PMID:41616795	TRPC6 inhibition for the treatment of focal segmental glomerulosclerosis: a randomised, placebo-controlled, phase 2 trial of BI 764198.	★★★	2026	Trachtman H
PMID:40455255	Phase 1 trials of BI 764198, a transient receptor potential channel 6 inhibitor, in healthy volunteers and participants with kidney impairment.	★★★	2025	Schultz A
PMID:40388293	Genotype-phenotype correlations and clinical outcomes of genetic TRPC6 podocytopathies.	★★★	2025	McAnallen SM
PMID:39352759	Natural History and Clinicopathological Associations of TRPC6-Associated Podocytopathy.	★★★	2025	Wooden B
PMID:36750145	The small GTPase regulatory protein Rac1 drives podocyte injury independent of cationic channel protein TRPC5.	★★★	2023	Polat OK
PMID:36007181	Ion channels and channelopathies in glomeruli.	★★★	2023	Staruschenko A
PMID:37615749	An inactivating human TRPC6 channel mutation without focal segmental glomerulosclerosis.	★★★	2023	Batool L

… and 85 more — see full_report.md

Empirical (32)

OpenTargets (1)

Source	Detail	Quality
OpenTargets · TRPC6	Open Targets association ENSG00000137672 / MONDO_0100313: overall=0.780, genetic=0.891, literature=0.812, affected_pathway=0.000, animal_model=0.000, clinical=0.000, somatic_mutations=0.000.	★★★

Genetics (7)

Source	Detail	Quality
ot_genetics_coloc:locus_breadth_summary	TRPC6 coloc: 23 distinct GWAS traits found (0 matched Focal Segmental Glomerulosclerosis). Excluded as off-indication: Colorectal cancer or advanced adenoma, Educational attainment, Menarche (age at onset), Height, Age when periods started menarche.	★★★
spoke:DOID:1312	SPOKE graph: TRPC6–focal segmental glomerulosclerosis association via DISEASES; gwas_p=n/a, diseases_score=6.386.	★★★
ontology:TRPC6	Ontology constraints for TRPC6: Mode of inheritance: Autosomal dominant (source: HPO/Monarch); HPO phenotype breadth: 7 phenotype(s) (moderate); top terms: Autosomal dominant inheritance, Chronic kidney disease, Hypertension, Focal segmental glomerulosclerosis, Proteinuria.	★★★
gwas_catalog:locus_breadth_summary	TRPC6 locus: 37 distinct GWAS traits found (0 matched Focal Segmental Glomerulosclerosis). Excluded as off-indication: Bone mineral density mean, MMP8/MMP9 protein level ratio, Brain rsFMRI measurement (amplitude of low frequency fluctations) (Net100_Node38), Height, Menarche (age at onset).	★★★
gencc:TRPC6	GenCC gene-disease validity for TRPC6: focal segmental glomerulosclerosis 2 (Strong, Ambry Genetics); focal segmental glomerulosclerosis 2 (Strong, Labcorp Genetics (formerly Invitae)); focal segmental glomerulosclerosis 2 (Strong, PanelApp Australia); familial idiopathic steroid-resistant nephrotic syndrome (Supportive, Orphanet).	★★★
orphanet:TRPC6 (2)	Orphanet associations for TRPC6: Hereditary steroid-resistant nephrotic syndrome (ORPHA:656; Disease-causing germline mutation(s) (gain of function) in; Assessed).; Orphanet prevalence: Hereditary steroid-resistant nephrotic syndrome (ORPHA:656): Unknown [Point prevalence, Worldwide, Not yet validated].	★★★

Expression & Omics (17)

Source	Detail	Quality
GTEx/HPA (14)	**Lung: 23.2 TPM**, **Esophagus_Muscularis: 19.0 TPM**, **Thyroid: 12.0 TPM**, Esophagus_Gastroesophageal_Junction: 9.5 TPM, Adipose_Subcutaneous: 7.9 TPM, Kidney_Cortex: 1.2 TPM, Brain_Cortex: 0.7 TPM, Brain_Cerebellum: 0.6 TPM, Pancreas: 0.3 TPM, Heart_Atrial_Appendage: 0.3 TPM, Heart_Left_Ventricle: 0.2 TPM, Whole_Blood: 0.1 TPM, Liver: 0.0 TPM, GTEx top tissues (median TPM): Lung=23.2, Esophagus_Muscularis=19.0, Thyroid=12.0, Esophagus_Gastroesophageal_Junction=9.5, Adipose_Subcutaneous=7.9. HPA specificity: Tissue enhanced. Subcellular: Cell membrane.	★★★

Source	Detail	Quality
spoke_anatomy:TRPC6	SPOKE knowledge graph confirms TRPC6 expression in 23 anatomical structure(s) (UBERON): Brodmann (1909) area 9, adipose tissue, adrenal gland, adult mammalian kidney, brain, calcaneal tendon, colon, corpus callosum, dorsolateral prefrontal cortex, frontal cortex (+13 more).	★★★
GTEx/HPA	TRPC6 RNA tissue specificity: Tissue enhanced (Human Protein Atlas).	★★★
UniProt	TRPC6 subcellular localization: Cell membrane (UniProt/HPA).	★★★

Constraint (3)

Source	Detail	Quality
gnomAD (3)	ClinVar variants in TRPC6 (via gnomAD): 40 Pathogenic (30 missense_variant, 5 synonymous_variant, 2 frameshift_variant, 2 stop_gained, 1 splice_acceptor_variant), 29 Likely Pathogenic (14 frameshift_variant, 9 missense_variant, 5 stop_gained, 1 splice_acceptor_variant), 124 Benign/Likely Benign out of 501 total.; gnomAD v4: 194 HC pLoF variants observed in TRPC6. Most common AF=2.63e-05. No homozygous carriers in gnomAD. Absence at this allele count is uninformative — not evidence of biallelic lethality; selection signal comes from LOEUF/o-e.; Constraint interpretation for TRPC6: LOEUF=0.759 (relatively LoF-tolerant (NOT haploinsufficient)); mis_z=1.70 (no meaningful missense constraint; note: higher mis_z = more missense-constrained); pLI=0.0000 (pLI-tolerant (consistent with LoF tolerance)); missense OEUF=0.928 (missense-tolerant (gene accepts amino-acid change globally)). syn_z=-0.000 (≈0 expected; |syn_z|>2 flags data quality).	★★★

Functional Genomics (2)

Source	Detail	Quality
DepMap	DepMap: TRPC6 dependent in unknown lines (Chronos_Combined).	★★★
project_score:TRPC6	Project Score: TRPC6 a fitness gene in 2/512 Sanger cell lines (scaled BF < 0). Mean scaled BF: 3.485 (negative = depletion).	★★★

Clinical Trials (2)

Source	Detail	Quality
NCT07220083	A Study to Find Out if BI 764198 Helps Adults and Adolescents With a Kidney Condition Called Focal Segmental Glomerulosclerosis (FSGS)	★★★
NCT05213624	A Study to Test BI 764198 in People With a Type of Kidney Disease Called Focal Segmental Glomerulosclerosis	★★★

▶ Dropped evidence (72 items)

Discovery — Analysis by Lens

✅ Genetics Lens — Support (confidence 90%)

> Genetic evidence strongly supports TRPC6 as a causal driver of focal segmental glomerulosclerosis (FSGS), with multiple pathogenic and likely pathogenic variants, strong Mendelian validation, and

> consistent missense-predominant spectrum in a LoF-tolerant gene.

The genetic evidence for TRPC6 as a causal driver of FSGS is robust. ClinVar reports 69 pathogenic or likely pathogenic (P/LP) variants, including 40 P and 29 LP, which strongly support Mendelian causality. The OpenTargets genetic association score of 0.891 further corroborates this gene-disease relationship. While no direct GWAS associations were found for FSGS, the absence is expected given the rarity of causal variants in Mendelian diseases. SPOKE graph supports a DISEASES edge with a high score of 6.386, reinforcing the genetic evidence independently. The constraint profile indicates that TRPC6 is relatively LoF-tolerant (LOEUF=0.759), consistent with its role as a gain-of-function disease gene. Missense variants predominate in this LoF-tolerant gene, with 37/62 missense P/LP variants compared to 25 truncating ones, supporting the gain-of-function mechanism. The high prevalence of TRPC6 mutations in FSGS patients (10-17%) further strengthens the causality argument. [⚠ MENDELIAN FLOOR GUARD: This text treats GWAS/colocalization absence as negative evidence, but Mendelian-grade causality support is already established (gold-star P/LP, ClinGen Definitive/Strong, and/or strong knowledge-graph corroboration). Absence of common-variant signal is EXPECTED for a Mendelian disease gene and must not lower causality.]

Axis	Verdict	Confidence	Rationale
Causality	Yes	90%	ClinVar reports 69 pathogenic or likely pathogenic variants, including 40 P and 29 LP, which strongly support Mendelian causality. The absence of GWAS signal is expected for a rare Mendelian disease gene.
[⚠ MENDELIAN FLOOR GUARD: This text treats GWAS/colocalization absence as negative evidence, but Mendelian-grade causality support is already established (gold-star P/LP, ClinGen Definitive/Strong, and/or strong knowledge-graph corroboration). Absence of common-variant signal is EXPECTED for a Mendelian disease gene and must not lower causality.]			
			Evidence: `1f443b6b…` , `ed89d2e5…` , `b838b896…`
Genetic Validity	Yes	90%	TRPC6 is relatively LoF-tolerant (LOEUF=0.759) and has a missense-predominant pathogenic spectrum, consistent with gain-of-function causality. The high prevalence of TRPC6 mutations in FSGS patients further supports the genetic validity.
			Evidence: `8a7af079…` , `bc7380bd…`

✅ Biology Lens — Support (confidence 90%)

> TRPC6 is strongly supported as a mechanistically understood and druggable target for FSGS, with multiple lines of evidence supporting its role in glomerular function and disease pathogenesis. The protein's subcellular localization to the cell membrane makes it amenable to both small-molecule and antibody-

> based therapies. CRISPR data indicates dependency across some cancer cell lines, which is not a concern for this non-oncology indication.

TRPC6 channels play a critical role in glomerular filtration and podocyte function, with multiple studies confirming their involvement in FSGS pathogenesis (e.g., TRPC6 mutations cause familial FSGS). The protein is expressed at high levels in the lung, esophagus muscularis, thyroid, and esophagus gastroesophageal junction, aligning with the disease-relevant tissue. Functional genomics data shows dependency across some cell lines, supporting its druggability. Small-molecule inhibitors targeting TRPC6 are being investigated for FSGS treatment, and antibody-based therapies may also be feasible given its membrane localization.

Axis	Verdict	Confidence	Rationale
Druggability	Yes	90%	TRPC6 has a known binding pocket (Open Targets) and is expressed in the cell membrane, making it amenable to both small-molecule and antibody-based therapies.
			Evidence: `c4a07737…` , `6346fe3d…`
Mechanism Of Action	Yes	90%	TRPC6 mutations are associated with FSGS, and gain-of-function studies demonstrate its role in podocyte injury (e.g., TRPC6-N110S mutation causes podocyte injury). CRISPR KO phenotypes show increased vasoconstriction and systemic arterial blood pressure, consistent with a functional role in the disease.
			Evidence: `3b9a793f…` , `55fe7dcc…`
Developability	Yes	90%	TRPC6 is expressed in the cell membrane, making it amenable to antibody-based therapies. Small-molecule inhibitors targeting TRPC6 are being investigated for FSGS treatment.
			Evidence: `3104ac86…` , `5e36ec93…`

✅ Clinical Lens — Support (confidence 80%)

> TRPC6 has strong clinical precedent with multiple Phase 2 and ongoing Phase 3 trials for focal segmental glomerulosclerosis (FSGS). BI 764198, a TRPC6 inhibitor, is being tested in these trials, with one completed Phase 2 trial showing proteinuria responses. The literature supports the role of TRPC6 mutations in causing FSGS and suggests that modulating this target could be therapeutically beneficial.

TRPC6 has been extensively studied both preclinically and clinically for its role in focal segmental glomerulosclerosis (FSGS). Multiple articles confirm the involvement of TRPC6 mutations in causing FSGS, with TRPC6 being upregulated in glomerular diseases. BI 764198 is a TRPC6 inhibitor that has shown proteinuria responses in a completed Phase 2 trial and is currently recruiting patients for a larger Phase 3 study. These clinical trials provide strong evidence of the therapeutic potential of targeting TRPC6 in FSGS.

Axis	Verdict	Confidence	Rationale
Clinical Precedent	Yes	90%	Multiple Phase 2 and ongoing Phase 3 trials are testing BI 764198 for FSGS, indicating strong clinical precedent.
			Evidence: `25dcaf76…` , `f53392a3…`
Clinical Validation	Yes	70%	BI 764198 showed proteinuria responses in a completed Phase 2 trial, providing evidence of therapeutic benefit.

Axis	Verdict	Confidence	Rationale
			Evidence: `f53392a3…`

✅ Safety Lens — Support (confidence 80%)

> The target TRPC6 is relatively LoF-tolerant and has multiple pathogenic variants associated with Focal Segmental Glomerulosclerosis, supporting its relevance to the disease. Mouse KO phenotypes include increased vasoconstriction and systemic arterial blood pressure, which are on-target liabilities in the kidney. Expression levels are high in lung, esophagus muscularis, thyroid, and esophagus gastroesophageal junction, raising concerns for on-target effects in these tissues. However, no serious adverse events or clinical trial data are reported, and the LOEUF indicates tolerance to haploinsufficiency. [⚠ CONSTRAINT GUARD: The text above mentions haploinsufficiency, but LOEUF=0.759 (relatively LoF-tolerant (NOT haploinsufficient)) does not support this — LOEUF ≥ 0.35 rules out haploinsufficiency. This claim should be removed.]

The target TRPC6 is relatively LoF-tolerant (LOEUF=0.759), which supports its safety for pharmacological inhibition. The presence of multiple pathogenic variants associated with Focal Segmental Glomerulosclerosis further reinforces the relevance of this target to the disease. Mouse KO phenotypes indicate increased vasoconstriction and systemic arterial blood pressure, suggesting potential on-target effects in the kidney. Expression levels are high in lung, esophagus muscularis, thyroid, and esophagus gastroesophageal junction, raising concerns for on-target effects in these tissues. However, no serious adverse events or clinical trial data are reported, and the LOEUF indicates tolerance to haploinsufficiency. The overall safety profile is acceptable with appropriate tissue expression. [⚠ CONSTRAINT GUARD: The text above mentions haploinsufficiency, but LOEUF=0.759 (relatively LoF-tolerant (NOT haploinsufficient)) does not support this — LOEUF ≥ 0.35 rules out haploinsufficiency. This claim should be removed.]

Axis	Verdict	Confidence	Rationale
Toxicity	Yes	80%	The target TRPC6 is relatively LoF-tolerant (LOEUF=0.759), which supports its safety for pharmacological inhibition. Mouse KO phenotypes indicate increased vasoconstriction and systemic arterial blood pressure, suggesting potential on-target effects in the kidney.
			Evidence: `8a7af079…`, `ed89d2e5…`, `b9b67eae…`
Tissue Specificity	No	80%	Expression levels are high in lung, esophagus muscularis, thyroid, and esophagus gastroesophageal junction, raising concerns for on-target effects in these tissues.
			Evidence: `143f268a…`, `f7d840b4…`, `61f64bcc…`, `890eb4a5…`

⚖️ Commercial Lens — Neutral (confidence 70%)

> The IP landscape is moderately encumbered by 12 patents naming TRPC6 as a target, though their scope and jurisdiction need further assessment. The competitive field is underserved at the target level but crowded in the broader indication space with one approved drug and multiple active clinical programs. The market size for hereditary steroid-resistant nephrotic syndrome, associated with Focal Segmental Glomerulosclerosis, could not be sized from Orphanet due to lack of validated prevalence data. [⚠ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1

> approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]

The IP landscape is moderately encumbered by 12 patents naming TRPC6 as a target. While the exact scope and jurisdiction of these patents are not detailed here, they suggest some level of patent protection that may need further analysis through an FTO study to ensure freedom to operate. The competitive field for this specific gene-targeting therapy is underserved; no approved or clinical-stage drugs targeting TRPC6 were found in the retrieved evidence. However, the broader indication space is crowded with one approved drug (SPARSENTAN) and multiple active clinical programs, indicating significant competition from other mechanisms of action. The market size for hereditary steroid-resistant nephrotic syndrome, a condition associated with Focal Segmental Glomerulosclerosis, could not be accurately sized due to the absence of validated Orphanet prevalence data. [⚠ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]

Axis	Verdict	Confidence	Rationale
Ip Landscape	No	80%	The IP landscape is moderately encumbered by 12 patents naming TRPC6 as a target, though their scope and jurisdiction need further assessment.
Competitive Opportunity	Yes	85%	The competitive field is underserved at the target level but crowded in the broader indication space with one approved drug and multiple active clinical programs.
[⚠ COMMERCIAL GUARD: 'underserved'/'uncrowded' applied to the indication, but the indication has 1 approved drug(s) and 38 active trial(s) — it is demonstrably contested. Scope any whitespace claim to the target.]			
Market Size	Uncertain	65%	The market size for hereditary steroid-resistant nephrotic syndrome, a condition associated with Focal Segmental Glomerulosclerosis, could not be accurately sized due to the absence of validated Orphanet prevalence data.

❓ Regulatory Lens — Insufficient evidence (confidence 0%)

> No evidence of this lens's type passed screening; verdict reflects the evidence gap, not a negative finding.

No FDA drug-label records were retrieved for TRPC6 in this run. This lens reasons only over FDA label data; an absence of records most often indicates that no approved drug targets this gene (a first-in-class signal) but cannot by itself confirm it. Disease-space regulatory context — approved therapies of other mechanisms, accepted endpoints, established trial pathways for the indication — is outside this lens's scope and is assessed by the clinical and commercial lenses. No label-level safety liabilities (black-box warnings, contraindications) can be inferred where no in-class approved drug exists.

Axis	Verdict	Confidence	Rationale

Cross-Lens Agreement

Consensus verdict: Support (confidence 85%) **Agreeing lenses:** biology, clinical, genetics, safety **Dissenting lenses:** commercial, regulatory

Target Suitability Score

TRPC6 — Score: 85/100

> TRPC6 is strongly supported by genetic and biological evidence for its role in FSGS, with multiple pathogenic variants and high expression levels in relevant tissues. Clinical trials have shown promising results with TRPC6 inhibitors, but the commercial landscape is moderately encumbered.

Gap Analysis

Literature (completeness 85%)

- No reviews or meta-analyses

Genetics (completeness 100%) - None identified

Clinical (completeness 75%)

- No large-scale clinical trials

Screening (completeness 100%) - None identified

Extraction (completeness 100%) - None identified

Lenses (completeness 60%)

- No specific lenses identified

Experiment (completeness 50%)

- Single experiment reported

Recommendations

PROCEED — Target shows strong validation across multiple lenses.

Supporting evidence:

- **Biology**: TRPC6 is strongly supported as a mechanistically understood and druggable target for FSGS, with multiple lines of evidence supporting its role in glomerular function and disease pathogenesis. The protein's subcellular localization to the cell membrane makes it amenable to both small-molecule and antibody-based therapies. CRISPR data indicates dependency across some cancer cell lines, which is not a concern for this non-oncology indication.
- **Clinical**: TRPC6 has strong clinical precedent with multiple Phase 2 and ongoing Phase 3 trials for focal segmental glomerulosclerosis (FSGS). BI 764198, a TRPC6 inhibitor, is being tested in these trials, with one completed Phase 2 trial showing proteinuria responses. The literature supports the role of TRPC6 mutations in causing FSGS and suggests that modulating this target could be therapeutically beneficial.
- **Genetics**: Genetic evidence strongly supports TRPC6 as a causal driver of focal segmental glomerulosclerosis (FSGS), with multiple pathogenic and likely pathogenic variants, strong Mendelian validation, and consistent missense-predominant spectrum in a LoF-tolerant gene.
- **Safety**: The target TRPC6 is relatively LoF-tolerant and has multiple pathogenic variants associated with Focal Segmental Glomerulosclerosis, supporting its relevance to the disease. Mouse KO phenotypes include increased vasoconstriction and systemic arterial blood pressure, which are on-target liabilities in the kidney. Expression levels are high in lung, esophagus muscularis, thyroid, and esophagus gastroesophageal junction, raising concerns for on-target effects in these tissues. However, no serious adverse events or clinical trial data are reported, and the LOEUF indicates tolerance to haploinsufficiency. [⚠ CONSTRAINT GUARD: The text above mentions haploinsufficiency, but LOEUF=0.759 (relatively LoF-tolerant (NOT haploinsufficient)) does not support this — LOEUF ≥ 0.35 rules out haploinsufficiency. This claim should be removed.]

Source Quality Notes

Quality assessments made on 131 evidence sources.